PROJECT REPORT

Water Reminder Application

Subject: Mobile Application Development

Platform: Android (Java/Kotlin)

Tool: Android Studio

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1. INTRODUCTION

The **Water Reminder Application** is an Android-based application designed to help users maintain proper hydration in their daily life. In today's fast-paced and busy lifestyle, individuals often neglect the fundamental necessity of drinking sufficient water, which can lead to various health complications such as dehydration, fatigue, and reduced cognitive function.

This application provides a seamless solution by empowering users to track their daily water consumption, configure personalized reminders, and monitor their long-term progress. Developed using Android Studio, the app emphasizes a clean, intuitive, and user-friendly interface to ensure a positive user experience.

2. AIM / OBJECTIVE

Aim:

To develop a robust mobile application that assists users in tracking their water intake and maintaining optimal hydration levels for better health.

Objectives:

- To design a simple and user-friendly interface that caters to all age groups.
- To allow users to log and track daily water intake accurately.
- To implement a notification system for timely hydration reminders.
- To provide a goal-oriented approach to help users achieve daily targets.
- To promote healthier lifestyle habits through consistent monitoring.

3. SYSTEM OVERVIEW

The Water Reminder App serves as a digital hydration coach. It utilizes local data management to provide a personalized experience without requiring constant internet connectivity.

The system architecture includes:

- **Welcome Screen:** The entry point of the application.
- **Target Input Screen:** Where users define their hydration goals.
- **Water Tracking Screen:** The main dashboard showing current progress.
- **Reminder System:** Logic to trigger system-level notifications.
- **Data Persistence:** Utilizes *SharedPreferences* for lightweight local storage of user targets and daily totals.

4. WORKING / LOGICAL FLOW

The application follows a linear and logical execution path:

1. **Initialization:** User opens the app and is greeted by the splash/welcome screen.
2. **Goal Setting:** User enters a daily water target (e.g., 3000 ml).
3. **Monitoring:** The home screen displays the current intake, starting at 0 ml every day.
4. **Interaction:** User clicks the “Add Water” button to log consumption (e.g., +250 ml).
5. **Automation:** The *AlarmManager* triggers notifications at set intervals to remind the user.
6. **Validation:** The system compares current intake with the target.
7. **Success:** A success message or visual indicator is shown once the target is reached.

5. FEATURES OF THE APPLICATION

- **Intake Tracking:** Real-time logging of water consumption.
- **Custom Targets:** Users can set goals based on their body weight or activity level.
- **Smart Notifications:** Push notifications that alert the user to drink water.
- **Progress Visuals:** Clear display of how much water is remaining for the day.
- **Lightweight Design:** Optimized to consume minimal system resources and battery.

6. MODULES OF THE APPLICATION

1. User Input Module: Handles the initial setup, accepting the daily target and saving it to the device's internal storage.

2. Tracking Module: The core logic that increments the water count and updates the UI instantly whenever the user adds a glass of water.

3. Reminder Module: Managed by the Android *AlarmManager* or *WorkManager* to ensure notifications are delivered even if the app is in the background.

4. Progress Module: A logical component that calculates the percentage of the goal achieved and provides feedback to the user.

7. REAL-LIFE USE CASES

- **Students:** Helps maintain focus and energy during long study sessions through proper hydration.
- **Office Professionals:** Reminds sedentary workers to take breaks and drink water regularly.
- **Fitness Enthusiasts:** Crucial for recovery and performance tracking during workouts.
- **Elderly Users:** Serves as a vital tool for those who may forget to drink water frequently.

8. ADVANTAGES

- **Health Benefits:** Reduces the risk of kidney stones, improves skin health, and boosts metabolism.
- **User Experience:** Minimalist design ensures no learning curve.
- **Privacy:** Data stays on the device, ensuring user privacy.
- **Efficiency:** Works completely offline without needing a data plan.

9. LIMITATIONS

- **UI Constraints:** The current version focuses on utility over advanced animations.
- **Data Sync:** Lack of cloud backup means data is lost if the app is uninstalled.
- **Platform:** Currently limited to Android devices only.

10. FUTURE SCOPE

- **Firebase Integration:** For cloud backup and cross-device synchronization.
- **Analytics:** Implementation of weekly and monthly graphs to track long-term trends.
- **AI Integration:** Reminders that adjust based on local weather conditions or physical activity.
- **Wearable Support:** Integration with smartwatches (Wear OS).

11. CONCLUSION

The Water Reminder Application successfully addresses a common health challenge using mobile technology. By providing a structured way to track and remind users about hydration, the app serves as a practical tool for lifestyle improvement. This project highlights the effectiveness of Android development in creating utility-based applications that have a direct positive impact on human health.

End of Project Report - Water Reminder App